Graph-level anomaly detection with normal-only training

First experiment: LGKDE, SIGNET and MUSE. LGKDE leads the shared 12-dataset comparison below. Also evaluate M-GLAD and HGOOD as recent candidates; their results come from different evaluation suites.

Scope and interpretation

One input is an entire graph; the output is a graph anomaly score. Eligible fitting uses normal graphs without real anomalous training examples. Synthetic perturbations derived from normal graphs are allowed. This is commonly called one-class, novelty detection, or semi-/partially unsupervised GLAD. It does not imply that every published tuning procedure also uses normal data only.

1 Comparable published ranking

Ordered by reported mean AUROC in LGKDE, Table 1 [1]. AUPRC comes from Table 5. Percentages; higher is better. The comparison uses 12 TU datasets and five trials. These are author-reported results, not new reproductions.

Rank	Method	AUROC	AUPRC	Official GitHub
1	LGKDE	79.37	84.01	MathAdventurer/LGKDE
2	SIGNET	77.34	69.92	yixinliu233/SIGNET
3	MUSE	76.63	71.93	kswoo97/GLAD_MUSE
4	UniFORM	76.58	71.03	Not verified
5	CVTGAD	76.03	73.77	jindongli-Ai/CVTGAD
6	GLADC	71.63	66.71	XuexiongLuoMQ/GLADC
7	OCGIN	71.34	70.28	LingxiaoShawn/GLOD-Issues
8	OCGTL	70.86	67.30	boschresearch / GLAD
9	GLocalKD	70.25	66.38	RongrongMa/GLocalKD

Coverage: MUTAG, PROTEINS, DD, ENZYMES, DHFR, BZR, COX2, AIDS, IMDB-BINARY, NCI1, COLLAB and REDDIT-BINARY. Ranking by mean AUROC differs from mean dataset rank or mean AUPRC. UniFORM is retained for completeness, but an official repository could not be verified. OCGIN's author-linked URL was not retrievable during this review.

2 Recent candidates and separate evidence

Do not compare the means in this table directly with page 1. The dataset sets, splits and experimental implementations differ. These numbers establish promising candidates, not a universal order.

Method / source	Reported performance	Official repository
M-GLAD IJCAI 2026 [2]	78.26 mean AUROC across 8 datasets (calculated). SIGNET: 74.46; OCGTL: 70.85 in the same table. M-GLAD wins 7/8 AUROC datasets.	Lab821-gz/M-GLAD Paper-linked; page not retrievable.
HGOOD IJCAI 2026 [3]	75.68 mean AUROC across 15 datasets (calculated from Table 2). AIDS: 99.23 +/- 0.76; REDDIT-B: 89.80 +/- 0.24.	Catfromuibe/HGOOD
HC-GLAD ArXiv v2 [4]	77.16 mean AUROC across 13 datasets (calculated from Table 2). AIDS: 99.51 +/- 0.38; REDDIT-B: 89.64 +/- 0.39.	Yali-Fu/HC-GLAD
GLADPro KDD 2025 [2, 8]	67.27 mean AUROC in the 8-dataset M-GLAD comparison (calculated). Useful when prototype explanations are a priority.	ZhenyuYangMQ/GLADPro
GOOD-D WSDM 2023 [3, 9]	72.70 mean AUROC on HGOOD's 15-dataset table (calculated). Established contrastive baseline; normal-only AD mode.	yixinliu233/G-OOD-D

Reproduction details that affect the choice

LGKDE: learns graph density using MMD and perturbed normal graphs. Set the repository's `mixed_anomaly_ratio` to zero for this setting. Validation-based early stopping is described, but the retrieved text does not establish a fully normal-only selection protocol [1, 6].

HGOOD: its official README confirms normal-only graph AD training. It also states that the supplied launch scripts are untuned and do not reproduce the fully tuned paper configuration by themselves [7].

HC-GLAD: explicitly defines normal-only training. The figures above are tied to the retrieved October 2024 preprint, not asserted to be the updated 2026 journal results. Several baseline figures are reused from GOOD-D [4].

OCGTL: official repository is archived. OCGIN also has an implementation in this repository, but that implementation belongs to the OCGTL authors [10].

3 How to use this ranking

Suggested comparison set: LGKDE, SIGNET, MUSE, CVTGAD, M-GLAD and HGOOD, plus OCGTL as an established one-class baseline. This shortlist is my synthesis of performance evidence, method diversity and code availability; it is not a statistically established global ranking.

For a strictly normal-only experiment

• Split normal graphs into training and validation sets before fitting. Keep real anomalous graphs exclusively in the final test set.

• Fit preprocessing, representations and reference sets using training normals. Use fixed epochs or a normal-only validation objective for model selection; never select checkpoints by test AUROC.

• Calibrate an operating threshold on held-out normals, for example their 99th anomaly-score percentile for a nominal 1% false-alarm rate. Also report AUROC and anomaly-positive average precision.

• Use identical graph splits, features, class direction, seeds and tuning budgets for every method. Report mean and standard deviation over repeated runs; do not infer significance from small mean differences.

Important evidence limits

UB-GOLD is a useful shared implementation [5], but its optimal-hyperparameter experiments explicitly search against test performance. Those scores are upper-bound comparisons, not evidence that no anomalous labels were used for selection. The main ranking here instead identifies its exact source as LGKDE; it still needs a tuning audit before strict deployment.

Class-derived TU anomalies may not represent operational failures, fraud, or rare molecular defects. AUPRC depends strongly on test prevalence. No experiments, installations or end-to-end repository reproductions were performed for this report. Public links and publications were checked; unavailable pages are identified above.

Sources and implementation links

[1] LGKDE, ICML 2026. Tables 1 and 5; Sections 3.1, 5.2 and Appendix E.2.

[2] M-GLAD, IJCAI 2026. Problem formulation and Table 1.

[3] HGOOD, IJCAI 2026. Table 2 (anomaly detection).

[4] HC-GLAD, arXiv:2407.02057v2. Sections 3.2, 5.1 and Table 2.

[5] UB-GOLD, ICLR 2025. Hyperparameter-search protocol, Section 3.3.

[6] LGKDE official repository and configuration documentation.

[7] HGOOD official README: normal-only mode and tuning caveat.

[8] GLADPro official repository, KDD 2025.

[9] GOOD-D official repository, WSDM 2023.

[10] OCGTL official repository, IJCAI 2022.